Flight Price EDA Analysis Report

Business Problem

Airline ticket pricing is influenced by multiple factors such as airline type, source and destination routes, number of stops, and flight duration. Customers often face fluctuating airfare prices, while airlines need better understanding of demand patterns and pricing behavior.

The objective of this project is to perform Exploratory Data Analysis (EDA) on flight booking data to identify: How flight prices vary with stops and duration. Which airlines dominate the market. How source and destination affect pricing. Patterns in ticket price distribution. Relationships between operational factors and airfare.

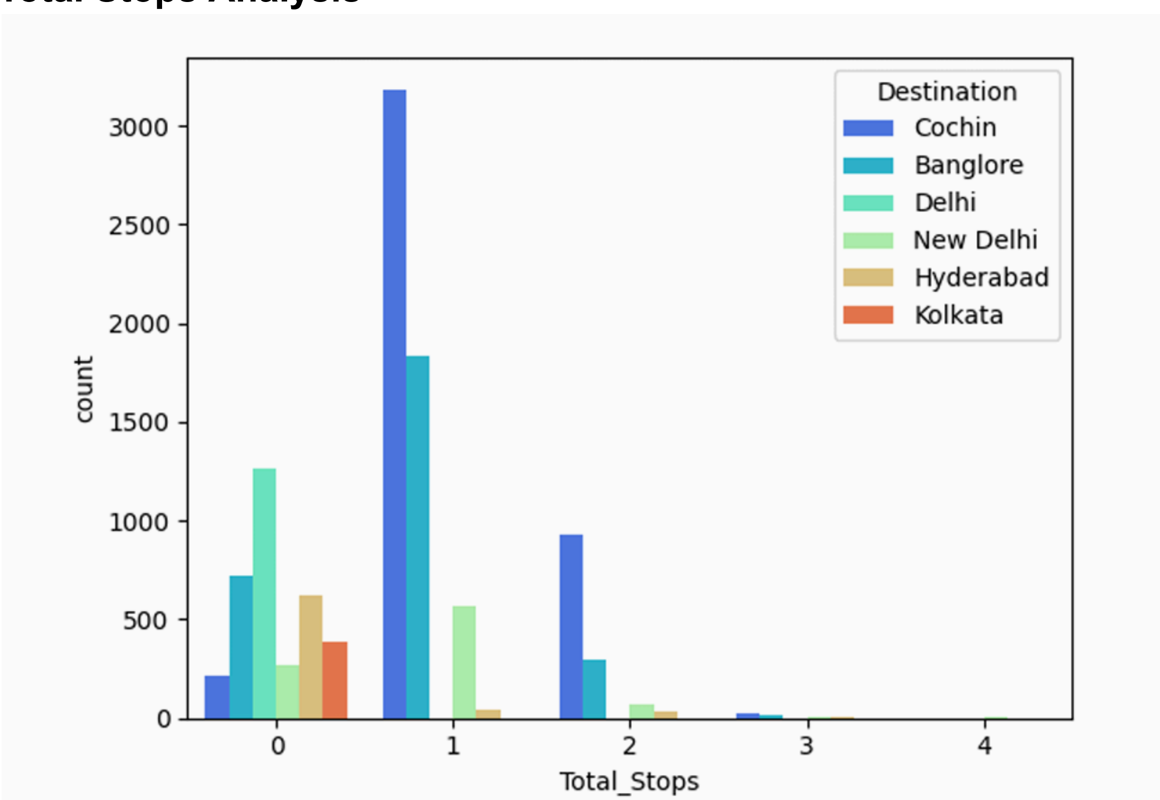
Hypothesis

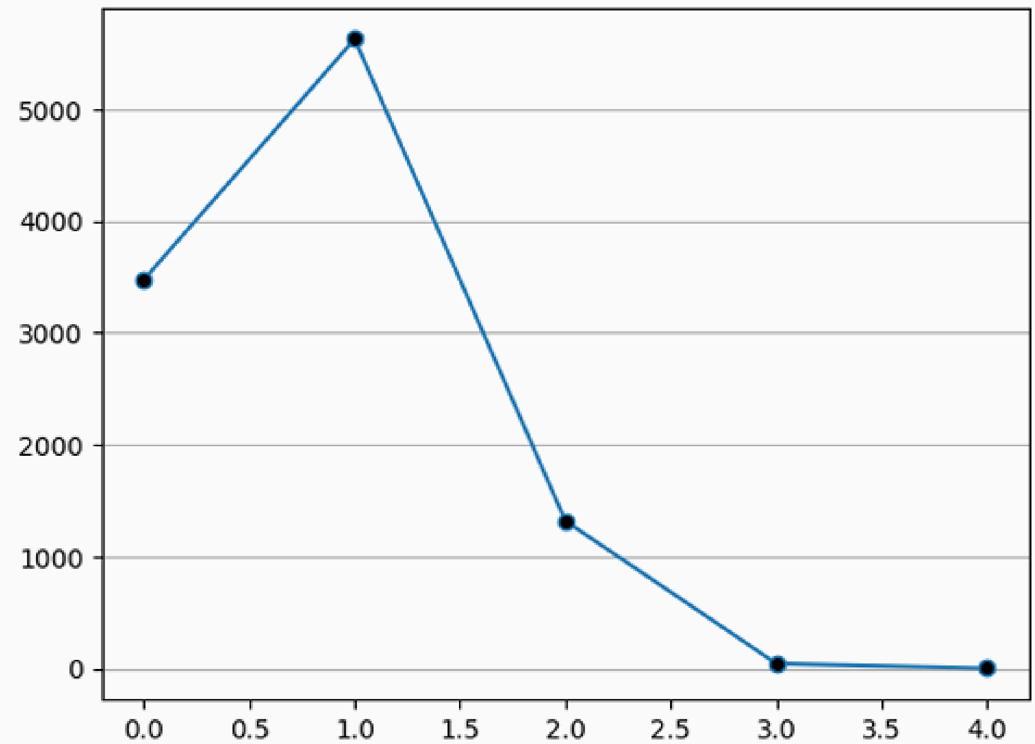
Flights with more stops generally have higher ticket prices. Longer duration flights tend to be more expensive. Popular routes such as Delhi to Cochin contribute major booking traffic. Airlines such as Jet Airways dominate the booking distribution. Ticket prices contain outliers due to premium airlines and business class travel.

Airline Distribution Analysis

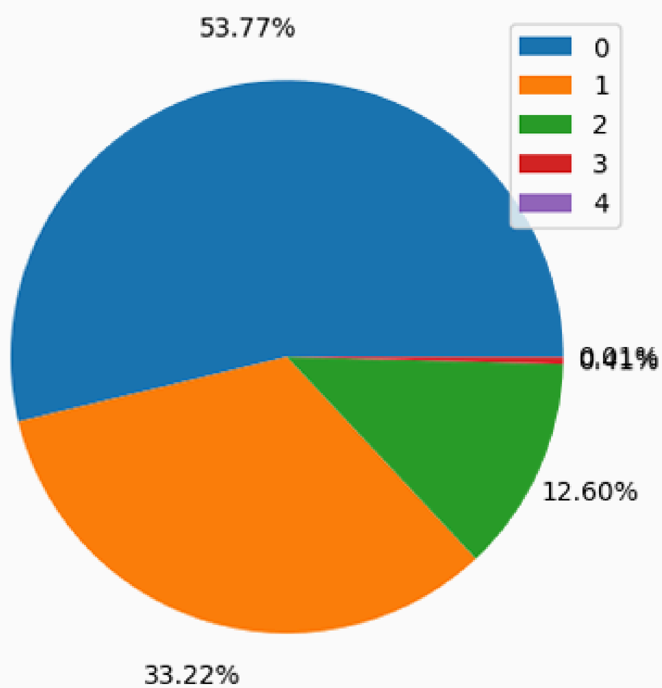
The visualization shows that Jet Airways, IndiGo, and Air India contribute the largest share of airline bookings. Jet Airways dominates the dataset, indicating higher market demand and passenger preference.

Total Stops Analysis






```
In [162…  wedges, texts, autotexts = plt.pie(df['Total_Stops'].value_counts(), autopct='
          plt.legend(wedges, df['Total_Stops'].value_counts().sort_index(ascending=True)
          plt.show()
```



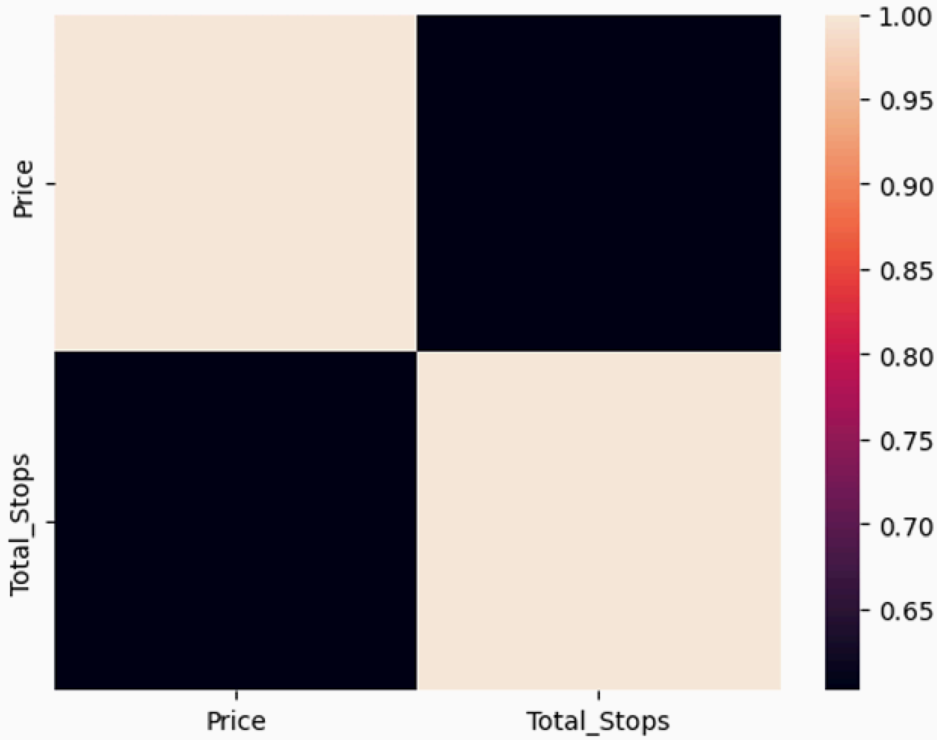
Most flights in the dataset contain either non-stop or single-stop journeys. Flights with higher numbers of stops are comparatively rare, indicating customer preference toward shorter travel routes.

Price vs Total Stops

The correlation analysis indicates a positive relationship between total stops and flight prices. Flights with more stoppages generally show higher airfare values due to longer routes and operational costs.

In [175…
```
sns.heatmap(map)
```

Out[175…
```
<Axes: >
```

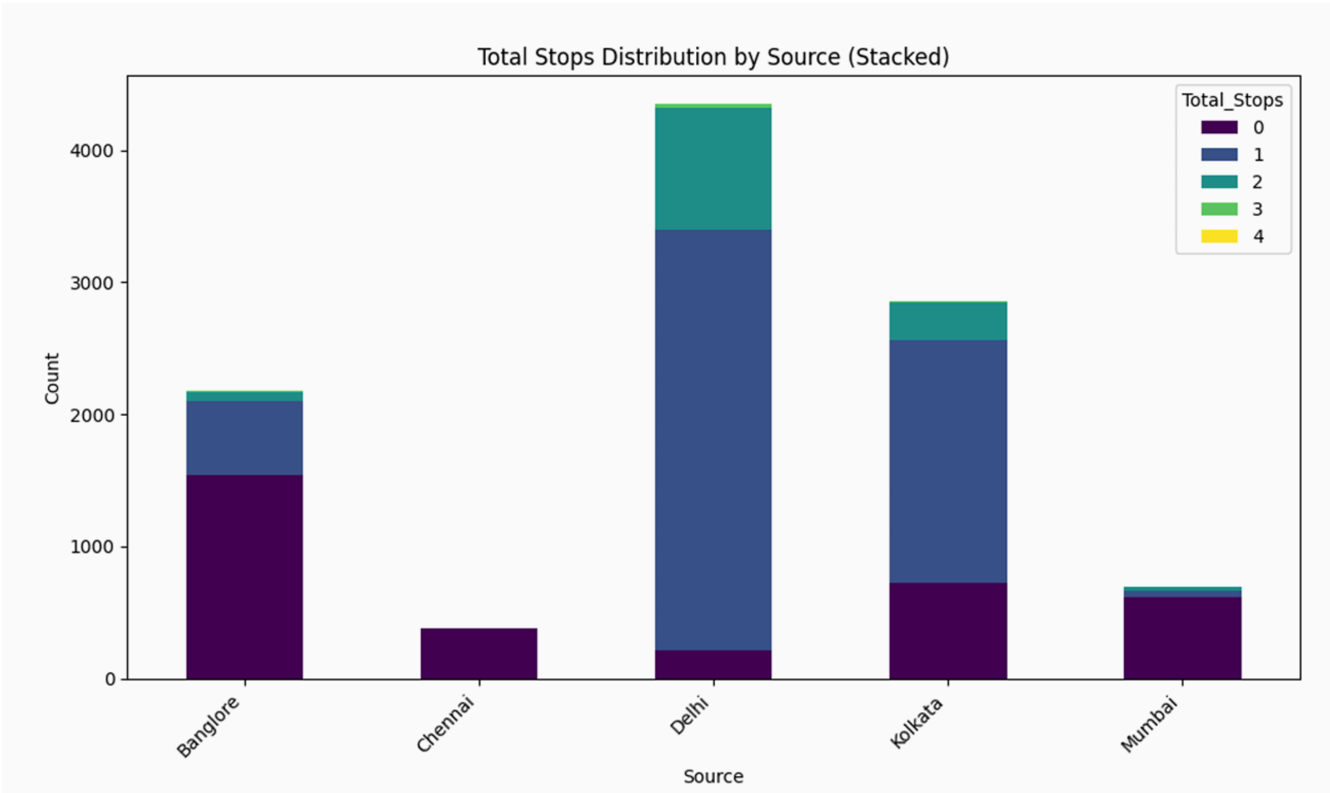


In [176…
```
df.describe()
```

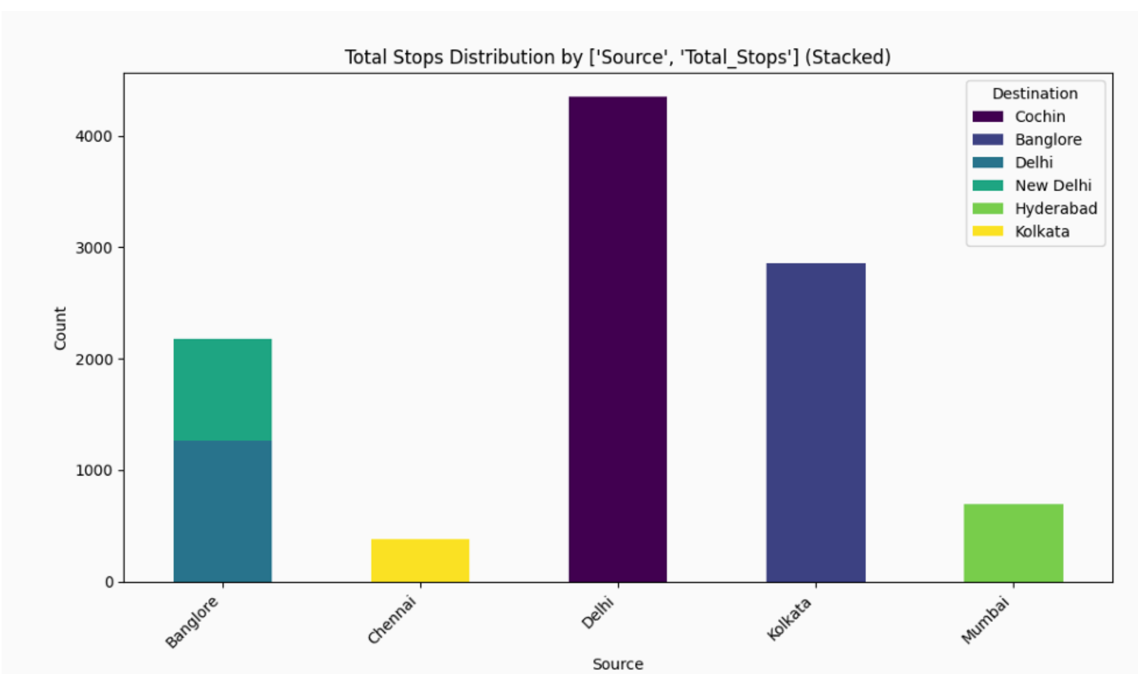
Out[176…

	Total_Stops	Price
count	10462.000000	10462.000000
mean	0.802332	9026.790289
std	0.660609	4624.849541
min	0.000000	1759.000000
25%	0.000000	5224.000000
50%	1.000000	8266.000000
75%	1.000000	12344.750000
max	4.000000	79512.000000

The graph indicates that Delhi is the busiest flight source in the dataset, with most flights having one stop. Kolkata also contributes a large number of flights, mainly with zero or one stop. Bangalore has many direct (non-stop) flights, while Chennai flights are almost entirely non-stop. Flights with three or four stops are very rare across all source cities, showing that passengers generally prefer shorter and more convenient routes with fewer stoppages.

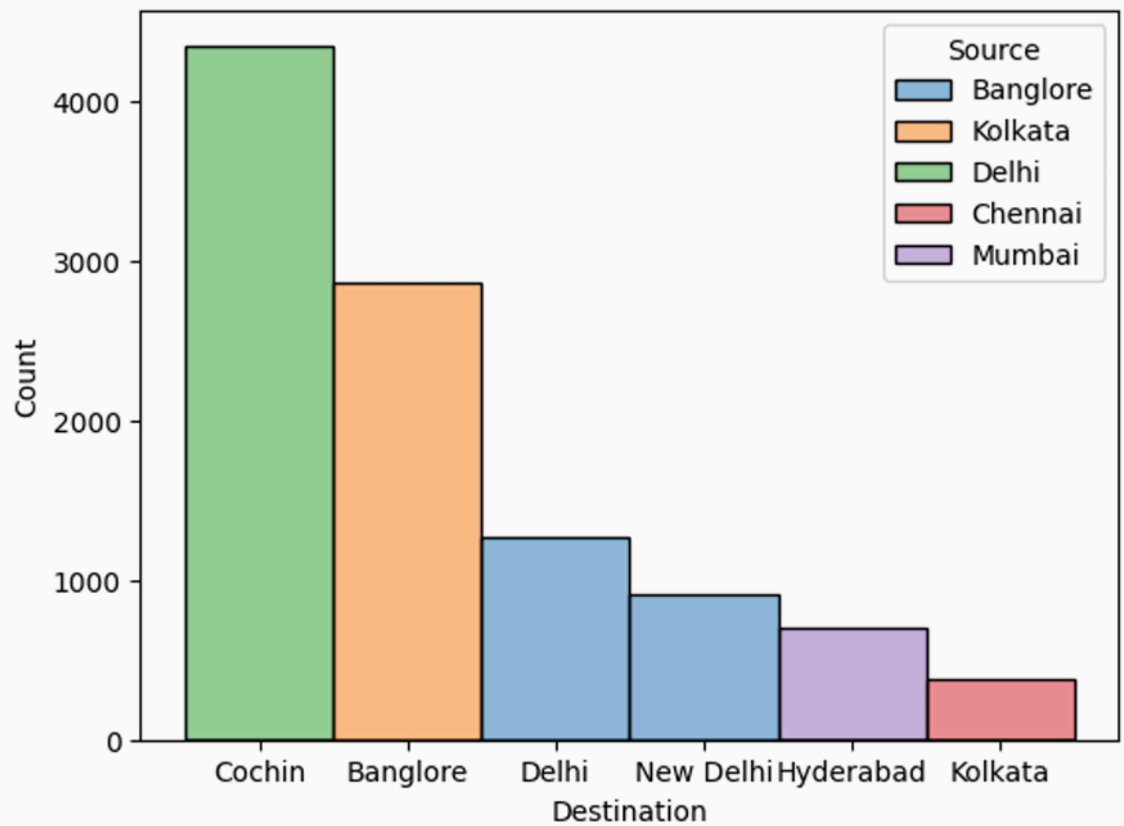


Source and Destination Trends



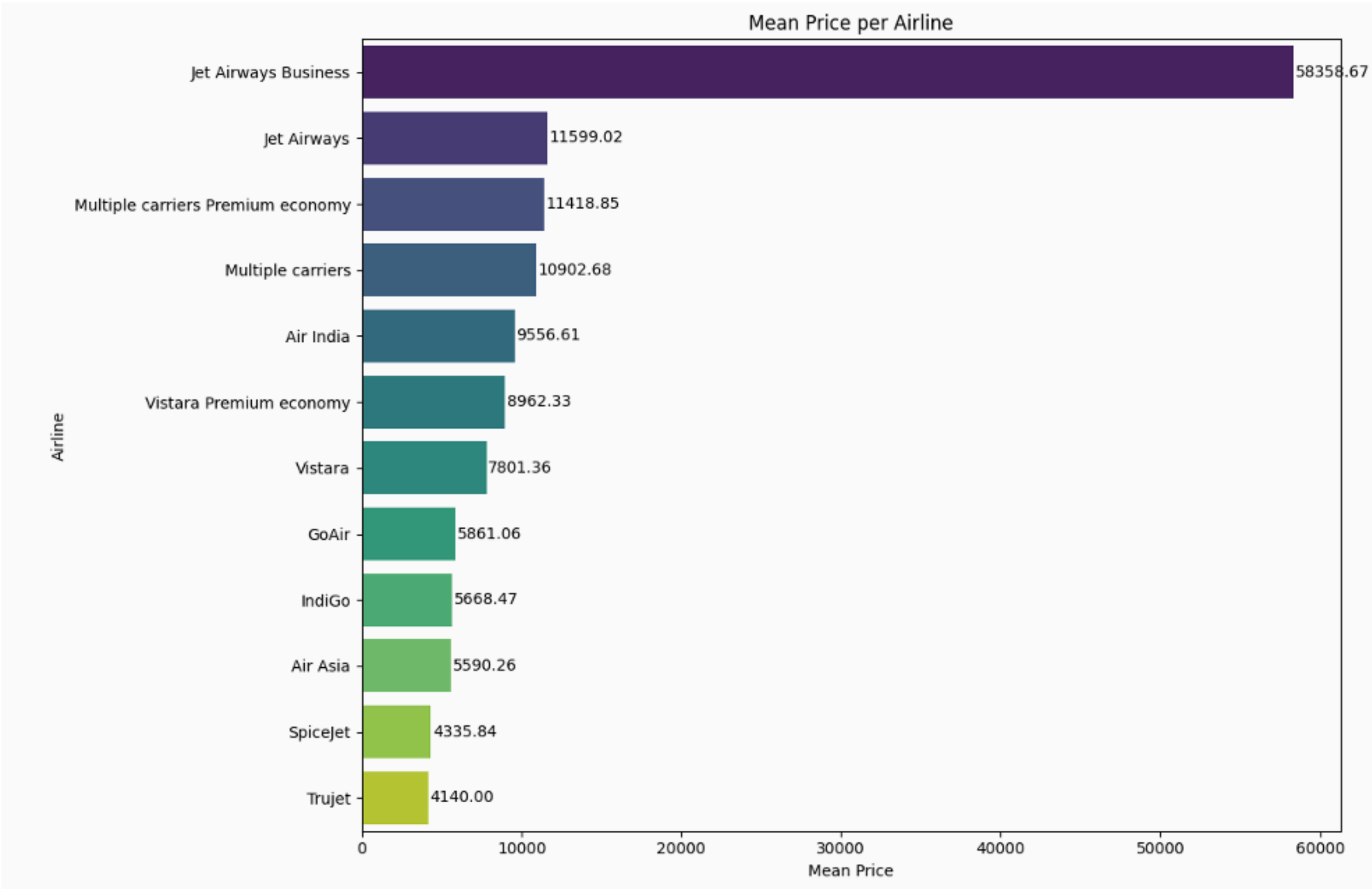
The graph shows that Delhi contributes the highest number of flights, mainly toward Cochin. Kolkata mostly connects to Bangalore, while Bangalore has flights primarily toward Delhi and New Delhi. Mumbai mainly connects to Hyderabad, and Chennai has comparatively fewer flights, mostly toward Kolkata. Overall, the chart highlights the dominant source-destination travel patterns in the dataset.

```
Out[184… <Axes: xlabel='Destination', ylabel='Count'>
```

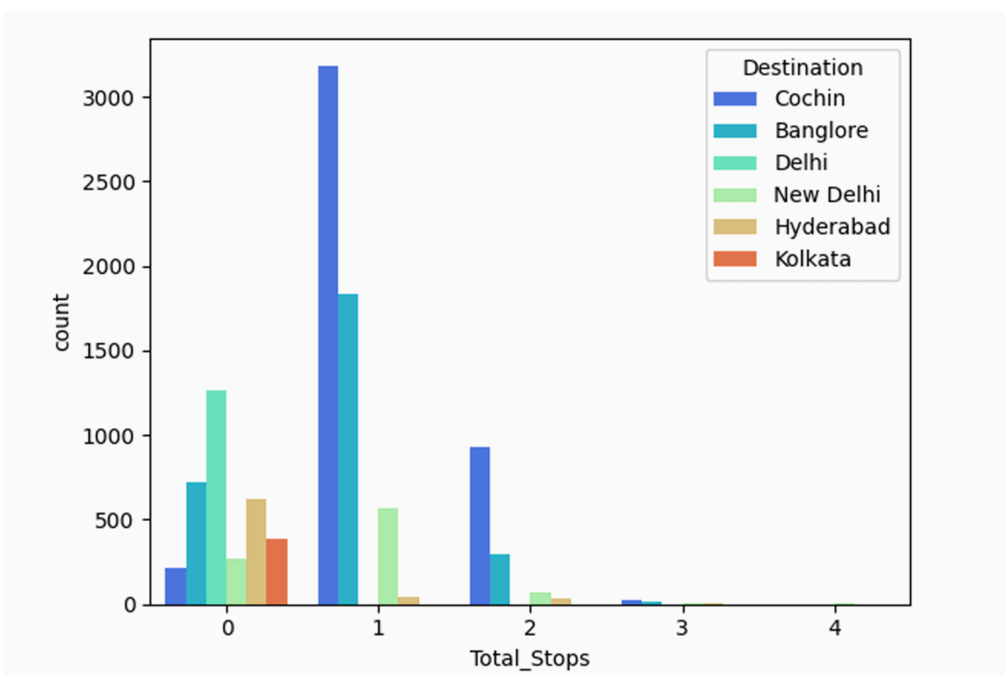


The graph indicates that Cochin is the most common destination, mainly receiving flights from Delhi. Bangalore is the second busiest destination with many flights from Kolkata. Hyderabad and Kolkata receive comparatively fewer flights. Overall, Delhi-to-Cochin is the busiest route in the dataset.

Average Airline Pricing



Premium airlines such as Jet Airways Business exhibit significantly higher average ticket prices. Budget airlines including IndiGo and SpiceJet maintain comparatively lower airfare ranges.
The graph shows that most flights have 1 stop, especially toward Cochin and Bangalore, while flights with more stops are rare and generally associated with higher ticket prices.



Model Training and Insights

A Random Forest Regression model was trained using airline, source, destination, stops, and duration features. The achieved R^2 score demonstrates moderate predictive capability for estimating flight prices.
Also trained a Linear Regression model which was giving lower accuracy than this, so dropped the idea of implementing that one.

```
preprocessor = ColumnTransformer(
    transformers=[
        ('ordinal', OrdinalEncoder(), ordinal_cols),
        ('nominal', OneHotEncoder(handle_unknown='ignore', sparse_output=False
        ('num', 'passthrough', numeric_cols)
    ]
)


# PIPELINE and MODEL INITIALISATION and TRIANING

model = RandomForestRegressor()

from sklearn.pipeline import Pipeline
from sklearn.linear_model import LinearRegression

model = Pipeline(steps=[
    ('preprocessing', preprocessor),
    ('regressor', RandomForestRegressor())
])

model.fit(X_train, y_train)

# EVALUATION
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error
import numpy as np

y_pred = model.predict(X_test)

print("R2 Score :", r2_score(y_test, y_pred))
print("MAE      :", mean_absolute_error(y_test, y_pred))
print("RMSE     :", np.sqrt(mean_squared_error(y_test, y_pred)))
```

```
R2 Score : 0.6563522898518892
MAE      : 1604.702341253883
RMSE     : 2361.83668782458
```

Suggestions

1. Airlines can optimize pricing strategies based on flight duration and stop patterns.
2. Dynamic pricing models should be applied on high-demand routes to maximize revenue.
3. Airlines should focus on reducing unnecessary stoppages to improve customer experience.
4. Machine learning models can be further improved using advanced feature engineering

Author -Ravi Namdeo